

Forest Fire Susceptibility – Bagmati Province, Nepal

A Machine-Learning Approach to Spatial Fire Risk

Forest Fire Susceptibility – Bagmati Province, Nepal

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Research Questions:

Susceptibility, not forecasting

Which areas of Bagmati Province face the highest fire risk, given vegetation, terrain, climate, and human pressure?

SUSCEPTIBILITY (our choice)

- Where is fire structurally likely?
- Static risk surface, p in $[0,1]$ per cell
- Label has no time axis
- → inputs averaged over time

Why This Matters

Why This Matters

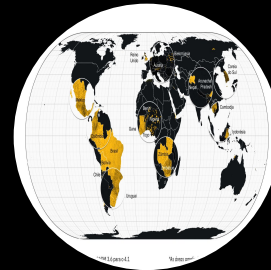
- Nepal's pre-monsoon fires (Feb–May) cause major annual damage
- April alone accounts for >43% of annual burned area
- ~65% of forested area in key landscapes falls in high-risk zones
- Climate change is intensifying dry-season fire risk
- Bagmati spans Chure → mid-hills → Himalaya: one rule can't fit it

(Mishra et al. 2022; Parajuli et al. 2020)

Study Area and Data Pipeline



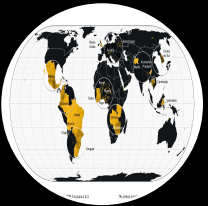
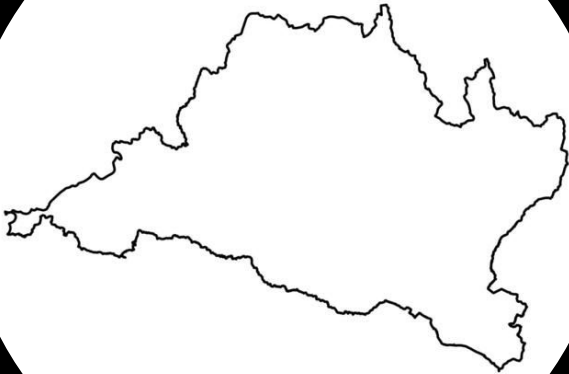
Study Area



Study Area

Bagmati Province, Nepal (27.0° – 28.4° N, 84.0° – 86.6° E)

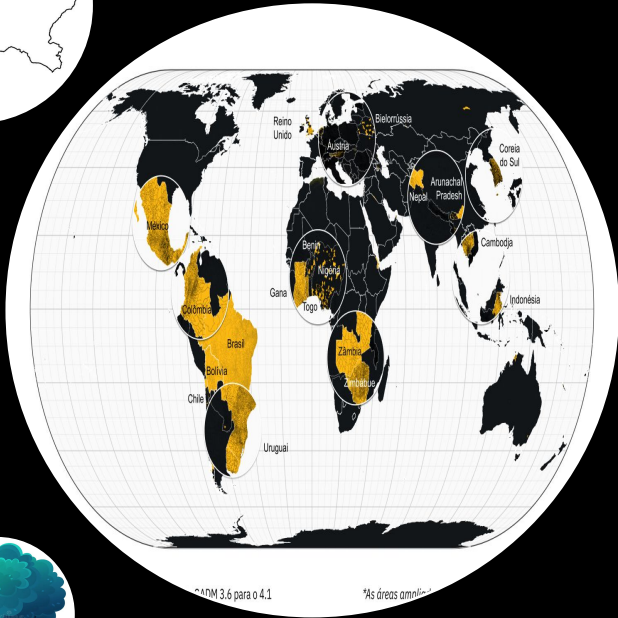
- Bounding box covers the full province extent
- Used to clip and align all raw datasets
- 13 districts; bbox 27.0° – 28.4° N, 84.0° – 86.6° E; $\sim 290 \times 155$ km
- Elevation: Chure hills $\rightarrow$ high Himalaya
- Fire season 1 Feb – 31 May; labels 2015–2025
- 4,131 observed fire cells, clustered in the mid-hills



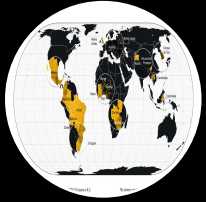
Study Area

13 districts, reconstructed
from GADM boundaries

- No "Bagmati" field in source data – built from district names
- Districts include Kathmandu, Lalitpur, Chitawan, Dolakha, etc.



Study Area



Spans Chure hills → mid-hills → high Himalaya

- Elevation ranges from lowland Chure to high mountains
- Creates strong gradients in temperature and rainfall



Study Area

Diverse vegetation, topography,
and human pressure

- Mix of dense forest, cropland, and urban settlement
- Ideal setting for a multi-factor fire risk model



Data Sources

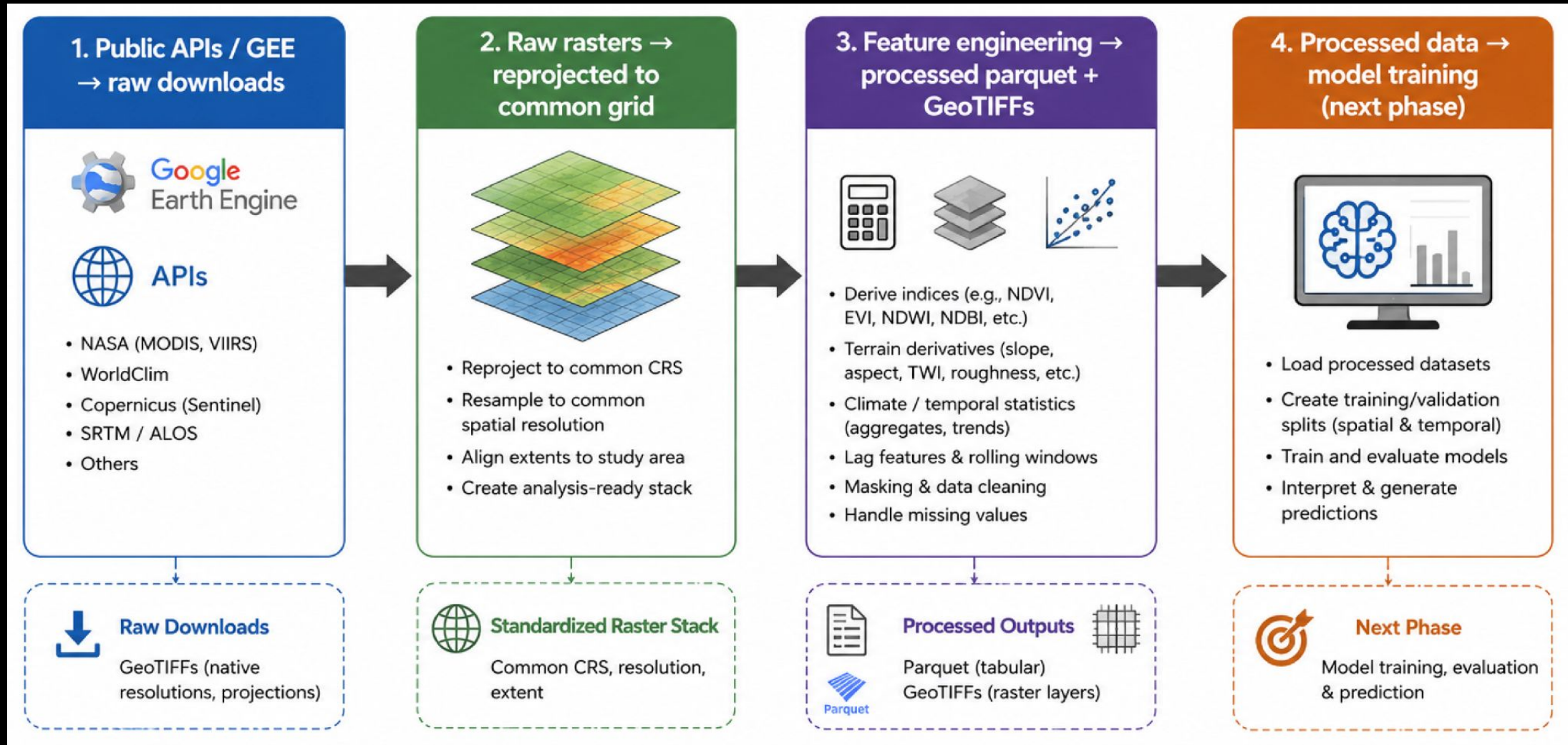
Eight public sources + Google Earth Engine

- NASA FIRMS – fire hotspot detections
- CGIAR SRTM – 30m elevation (DEM)
- ESA WorldCover – 10m land cover
- Open-Meteo – historical daily weather
- OSM Overpass – roads, settlements, water
- WorldPop – population density
- Hansen Global Forest Change – tree cover & loss
- Google Earth Engine – NDVI, LST, Sentinel-2, burned area (MCD64A1)

Source	Gives	Res
NASA FIRMS	fire labels	~1 km
DEM (SRTM/CopDEM)	terrain	30 m
ESA WorldCover	land cover	10 m
Open-Meteo	daily weather	point
OSM	roads/water	vector
WorldPop	population	100 m
Hansen GFC	tree cover	30 m
GEE NDVI/LST/S2	veg, heat, clim	250m-1km

Pipeline Architecture

Pipeline Architecture



Feature Engineering



Feature Engineering

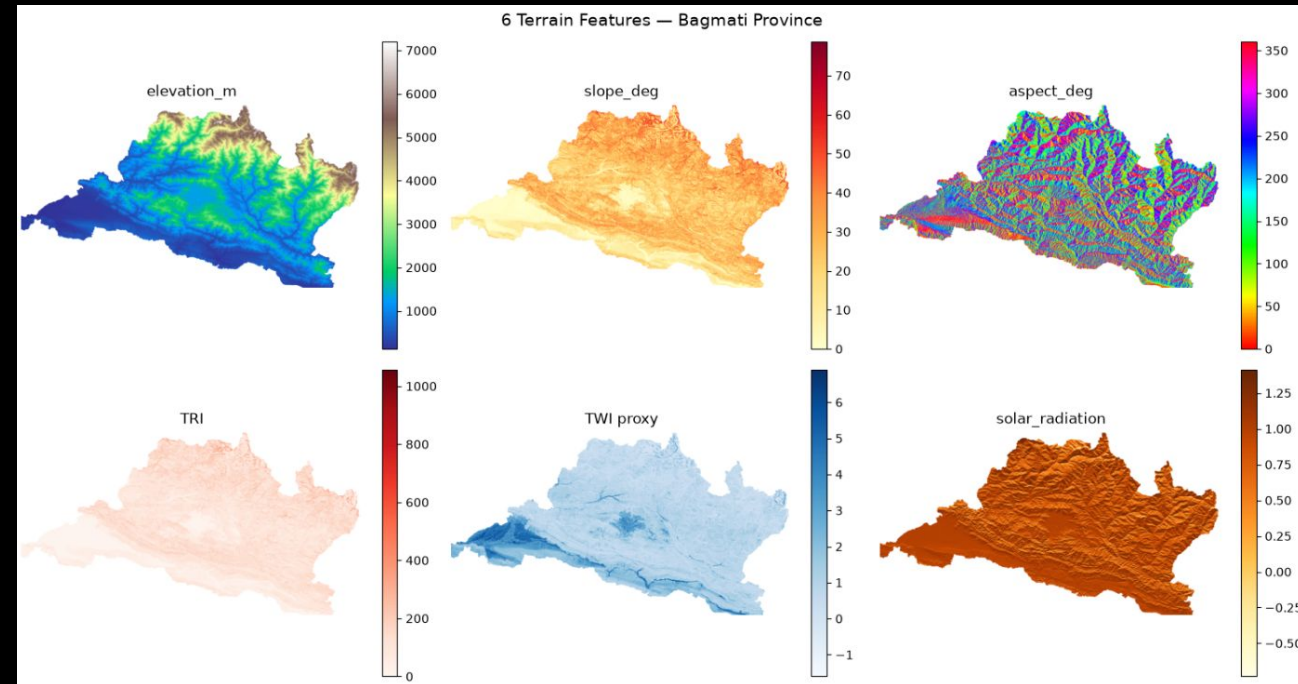
- 6 feature groups + 1 target variable
- Output: single tabular dataset, one row per grid cell
- Final exports: .parquet dataset + feature-stack
GeoTIFFs

Features

Features

Terrain Features

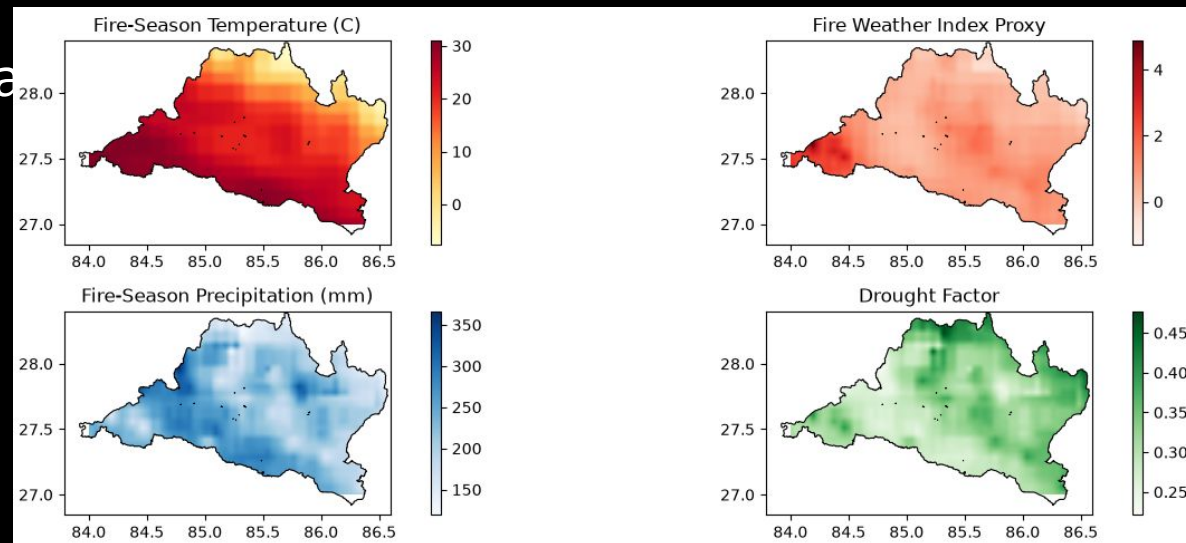
- `elevation_m` — from SRTM DEM
- `slope_deg` — computed from elevation gradient
- `aspect_deg` — slope direction ($0-360^\circ$)
- TRI — Terrain Ruggedness Index
- TWI — Topographic Wetness Index



Features

Climate Features

- Max mean temperature ($^{\circ}\text{C}$), fire season
- Max mean wind speed (km/h), fire season
- Fire-season rainfall (mm)
- Drought factor (dryness index)
- Fire Weather Index proxy
- Longest consecutive dry-day streak



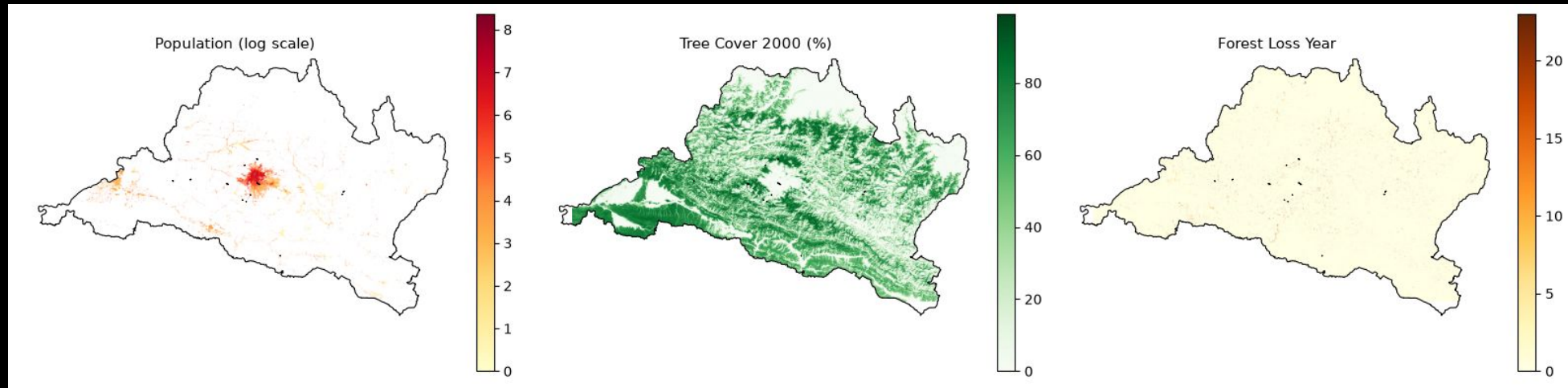
Features

Vegetation Features

- NDVI – annual, fire-season, pre-monsoon, peak, post-monsoon
- NDVI anomaly – pre-monsoon NDVI minus annual mean
- EVI, NBR, NDWI – additional spectral indices (Sentinel-2)
- Captures vegetation health and moisture stress

Features

Land Surface Temperature, Forest Cover & Population

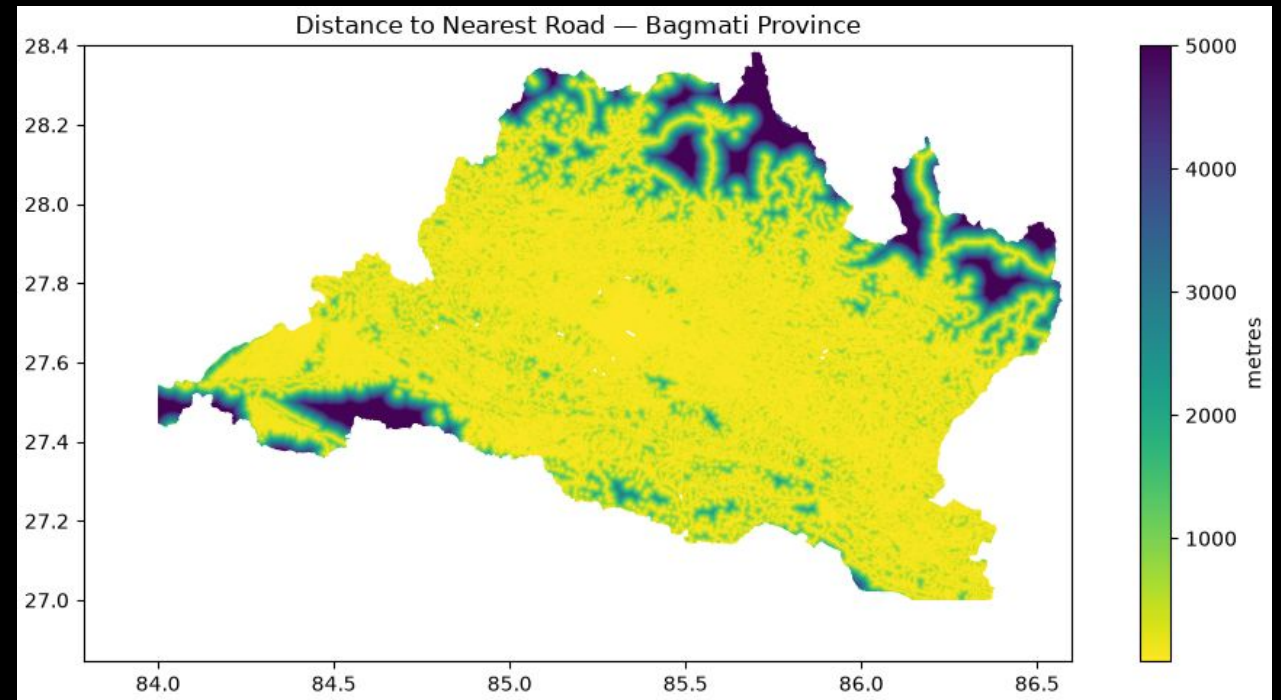


- LST – average fire-season land surface temperature
- Hansen tree cover (2000 baseline) + forest loss year
- WorldPop – 100m population density (2020)

Features

Distance / Human Proximity Features

- Distance to nearest road
- Distance to nearest settlement
- Distance to nearest water body
- Distance to nearest forest edge
- Distance to nearest cropland



Target Variable: Fire Label

Target Variable: Fire Label

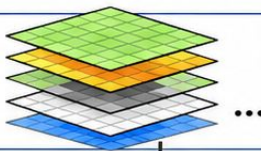
- Source: NASA FIRMS hotspot detections (2015–2024)
- `fire_occurred` – binary label (did a fire fall in this cell?)
- `fire_count` – number of detections per cell
- This is the variable the model will learn to predict

Final Dataset Assembly

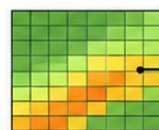
Final Dataset Assembly

1 Final Feature Rasters

(Climate, Terrain, NDVI, Distance Layers, etc.)



2 Flatten Raster Layers into 1D Pixel Records



pixel_id	f1	f2	...	f _n
1	0.23	15.2	...	0.71
2	0.25	14.8	...	0.66
...	...	...	...	...
N	0.31	16.1	...	0.80

3 Apply Masks

- Province Boundary
- Flammable Land Cover



4 Merge All Features into Single Tabular Dataset

(Rows = Pixels, Columns = Variables)

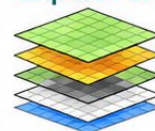
pixel_id	lat	lon	elev	slope	NDVI	dist_road	temp	...	fire
1	28.456	84.123	1345	18.3	0.62	312	22.1	...	0
2	28.457	84.124	1360	17.8	0.65	298	22.0	...	1
...	...	...	...	...	...	...	...	...	...
N	28.789	84.567	2105	26.1	0.41	523	18.6	...	0

5A Export Tabular Dataset



- ✓ Efficient storage
- ✓ Scalable
- ✓ Ready for ML

5B Export Per-Feature GeoTIFFs



- ✓ One GeoTIFF per feature
- ✓ Preserves spatial structure

6 Ready for Exploratory Analysis and Modelling



Explore data distributions



Identify patterns and relationships



Feature importance and diagnostics



Model training and prediction

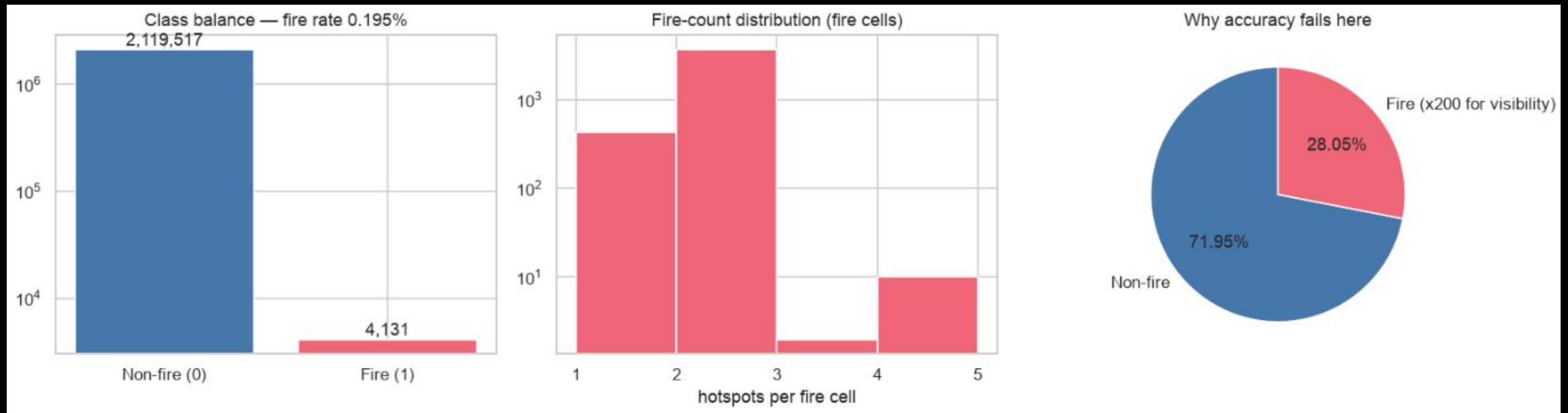
Exploratory Data Analysis

EDA – Dataset Overview

- 2,652,763 total grid cells in the bounding box
- 2,123,648 flammable cells retained for modelling
- 4,131 fire cells – only 0.195% of flammable area
- Class imbalance ratio: 513 : 1 (non-fire : fire)
- Accuracy alone is meaningless – use AUC-PR as primary metric

EDA – Class Imbalance & Spatial Distribution

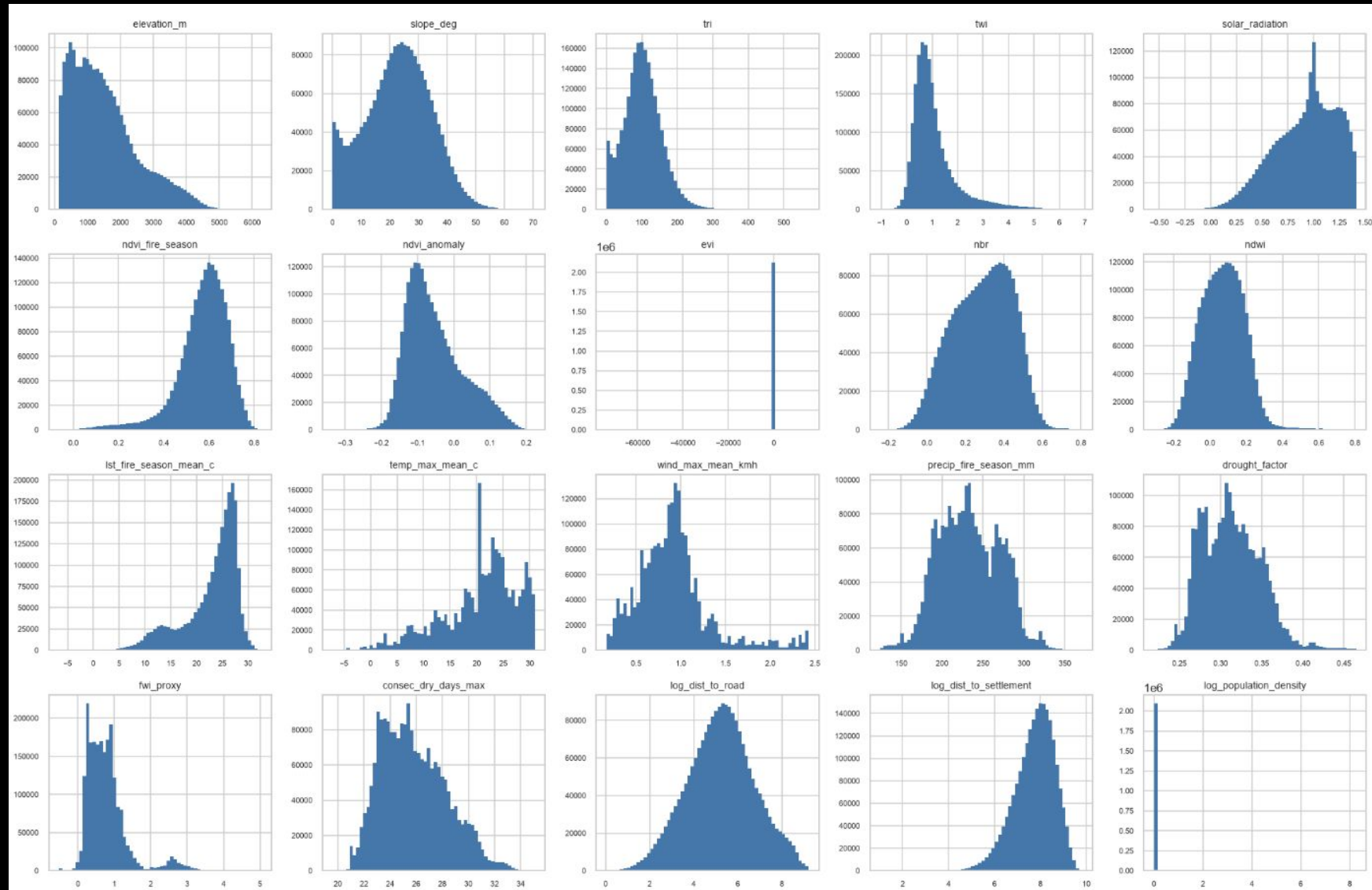
- Fire occurrence confirmed at **0.195%**
 - Predicting 'no fire' everywhere scores 99.8% accuracy – completely useless
- Fires **cluster spatially** – not randomly scattered
 - Concentrated in Chure belt, ridgelines, forest-cropland edges
- Clustering is why **random splits fail**
 - Neighboring pixels share near-identical features – leaks information to test set
- Spatial clustering justifies **block cross-validation** strategy



EDA – Feature Distributions & Correlations

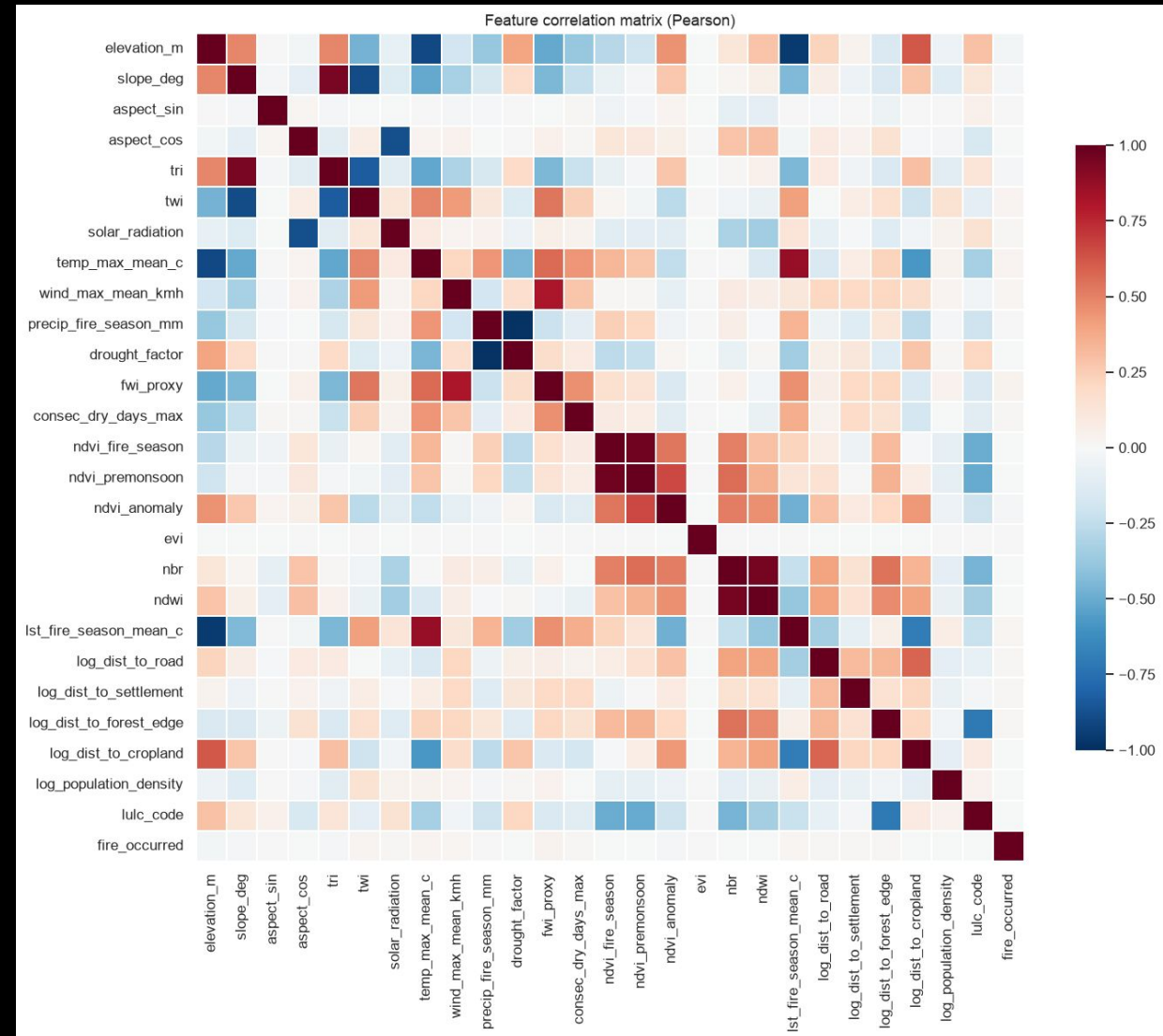
- Distance features and population density are **right-skewed**
 - Log-transformed before modelling to compress long tails
- Strongest fire vs non-fire separation on:
 - **slope_deg**, ndvi_premonsoon, lst_fire_season_mean_c
- High correlation among **NDVI variants**
 - Redundant variants dropped; NDVI anomaly added as derived stress feature
- Summary statistics computed for all **26 continuous features**

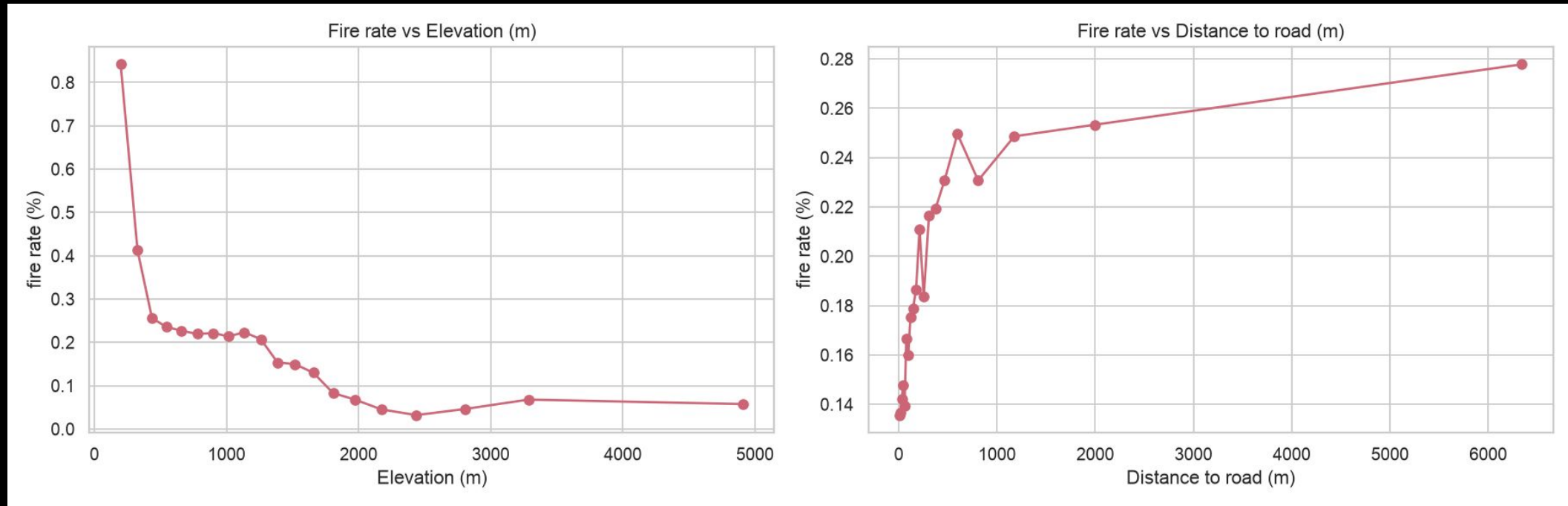
EDA – Feature Distributions (Visual Representation)



EDA – Correlations (Visual Representation)

```
Top correlations with fire_occurred:  
fwi_proxy          0.033  
twi                0.031  
elevation_m        0.027  
temp_max_mean_c    0.025  
lst_fire_season_mean_c 0.023  
wind_max_mean_kmh  0.022  
ndvi_anomaly       0.021  
slope_deg          0.020  
tri                0.019  
consec_dry_days_max 0.013  
Name: fire_occurred, dtype: float64
```





- Fire cells: **lower fire-season NDVI** (drier fuel)
- **Higher** surface temperature / LST
- **Closer** to roads and settlements
- Fire rate rises **monotonically** across drought / proximity bins

Model Training

Train / Test Strategy

- Spatial block CV: province divided into ~11 km tiles
 - 230 blocks total – 20% held out as test (46 blocks)
- Train: 1,680,339 rows | Test: 443,309 rows
- Fire cells in test set: 727
- 5-Groupfold spatial Block CV run on RF + XGBoost for robustness check
- Imputation + scaling rebuilt inside each fold – no leakage

Why spatial block cross-validation

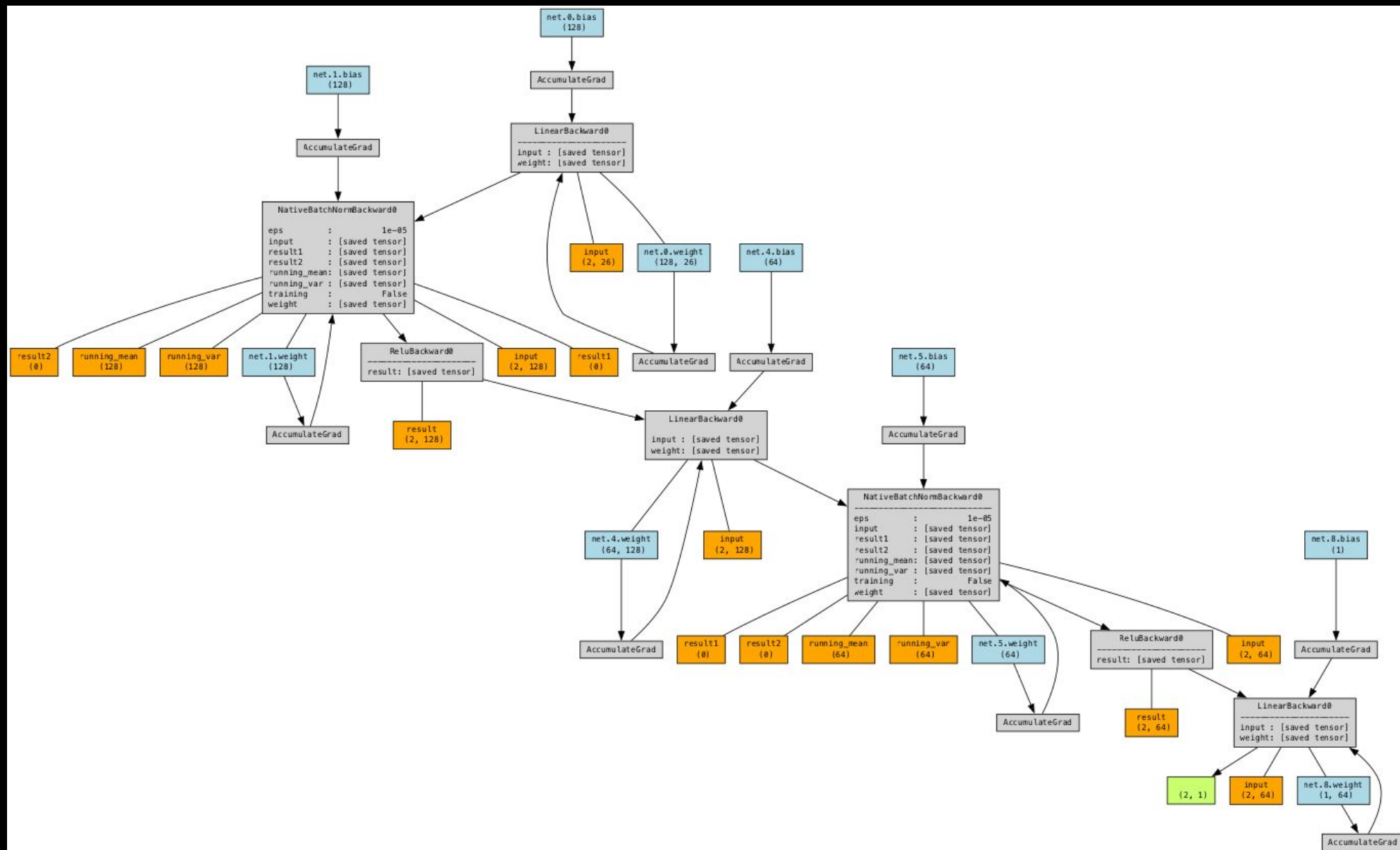
- Neighbouring cells are nearly identical (spatial autocorrelation)
- Random split → fire clusters leak into train and test
- → falsely optimistic scores (Roberts et al. 2017)
- Fix: tile into 0.1° (~11 km) blocks; hold out whole blocks
- 46 of 230 blocks (20%) held out as spatial test set
- Honest & pessimistic — generalisation to unseen terrain

Models Trained

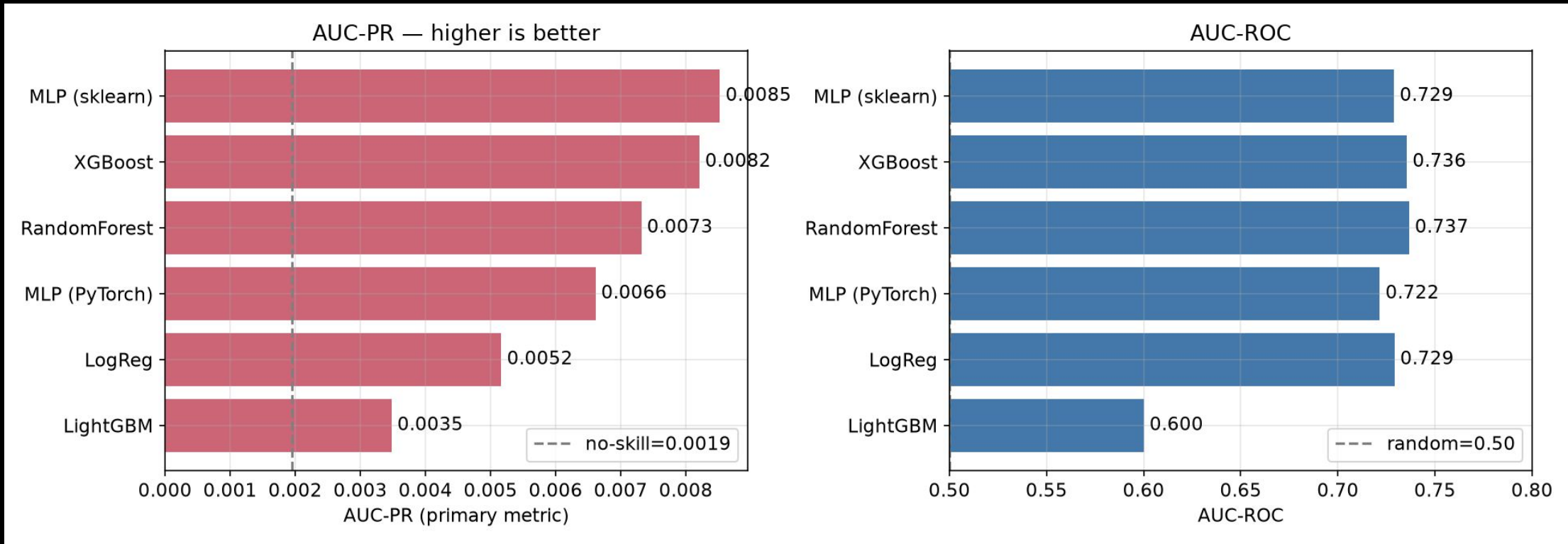
- **Logistic Regression** – interpretable baseline
- **Random Forest** – 300 trees, balanced class weights
- **XGBoost** – gradient boosting, scale_pos_weight for imbalance
- **LightGBM** – leaf-wise boosting, fast training
- **sklearn MLP** – 2-layer neural network
- **PyTorch MLP** – 128→64→1, weighted cross-entropy loss
- All models trained on **identical splits** and preprocessing

Model	Family
Logistic Regression	Linear (baseline)
Random Forest	Bagged trees
XGBoost	Boosted trees
LightGBM	Boosted trees
MLP (sklearn)	Neural net
MLP (PyTorch)	Neural net

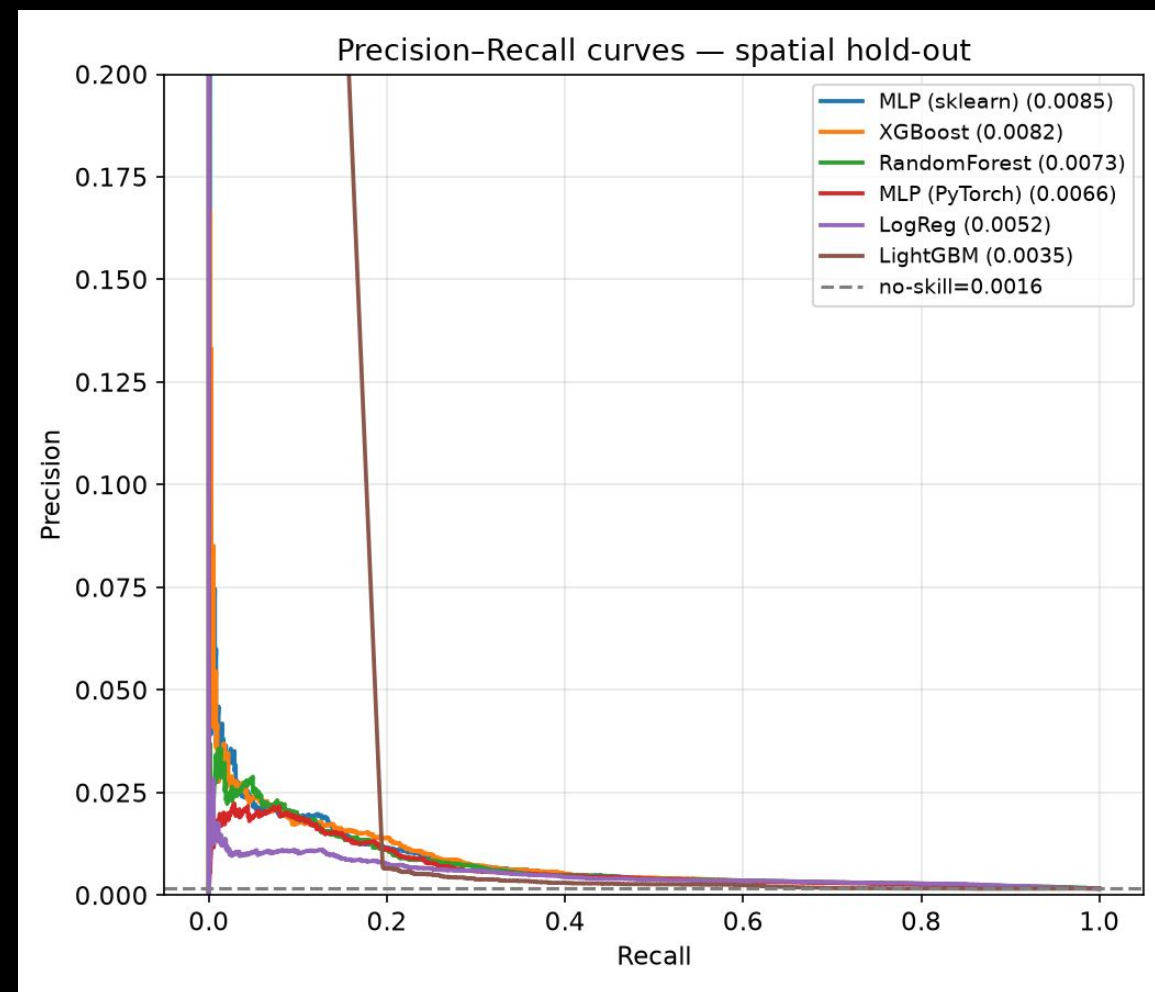
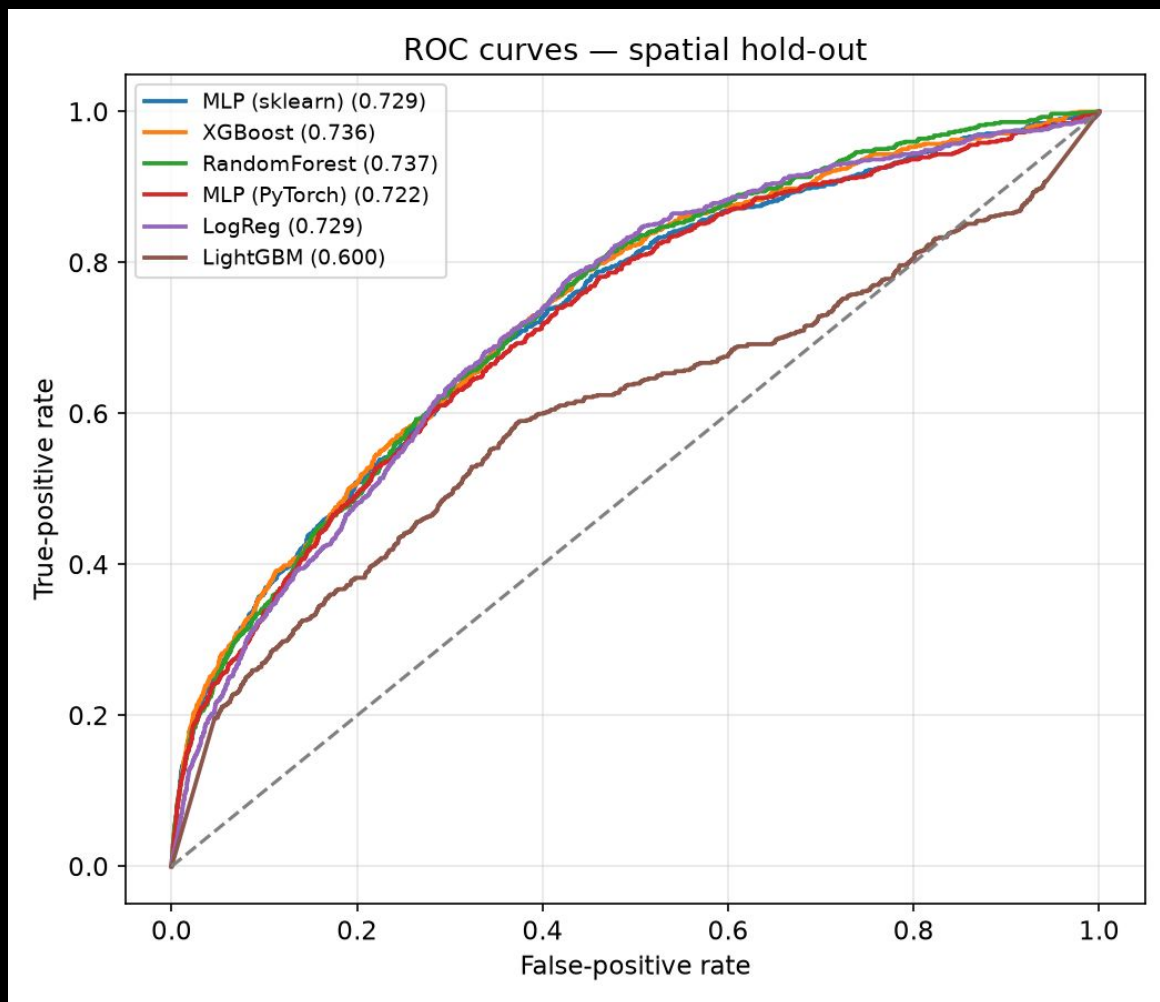
Model Architecture



Models Comparison



- 443,309 cells, **727 fires**; baseline AUC-PR **0.0016**
- All beat no-skill by **2–6x**; AUC-ROC **~0.73**
- Two MLPs ~ boosted trees; **LightGBM lags**
- Honest headline: **the map, not one model**

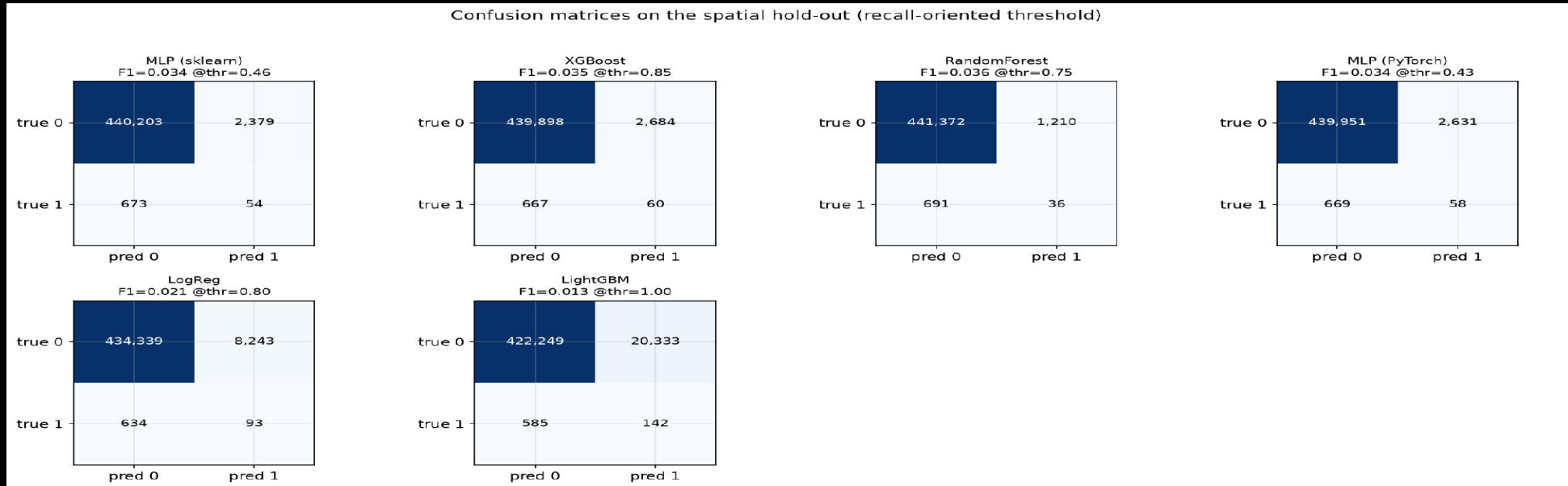


Model Comparison Results ➔

Model Comparison Results

- Primary metric: **AUC-PR** (precision-recall, handles imbalance)
- MLP (sklearn): AUC-PR = **0.0085** → best model
- XGBoost: AUC-PR = **0.0082**
- Random Forest: AUC-PR = **0.0073**
- Logistic Reg: AUC-PR = **0.0052**
- LightGBM: AUC-PR = **0.0035**
- AUC-ROC secondary – XGBoost leads at **0.7355**

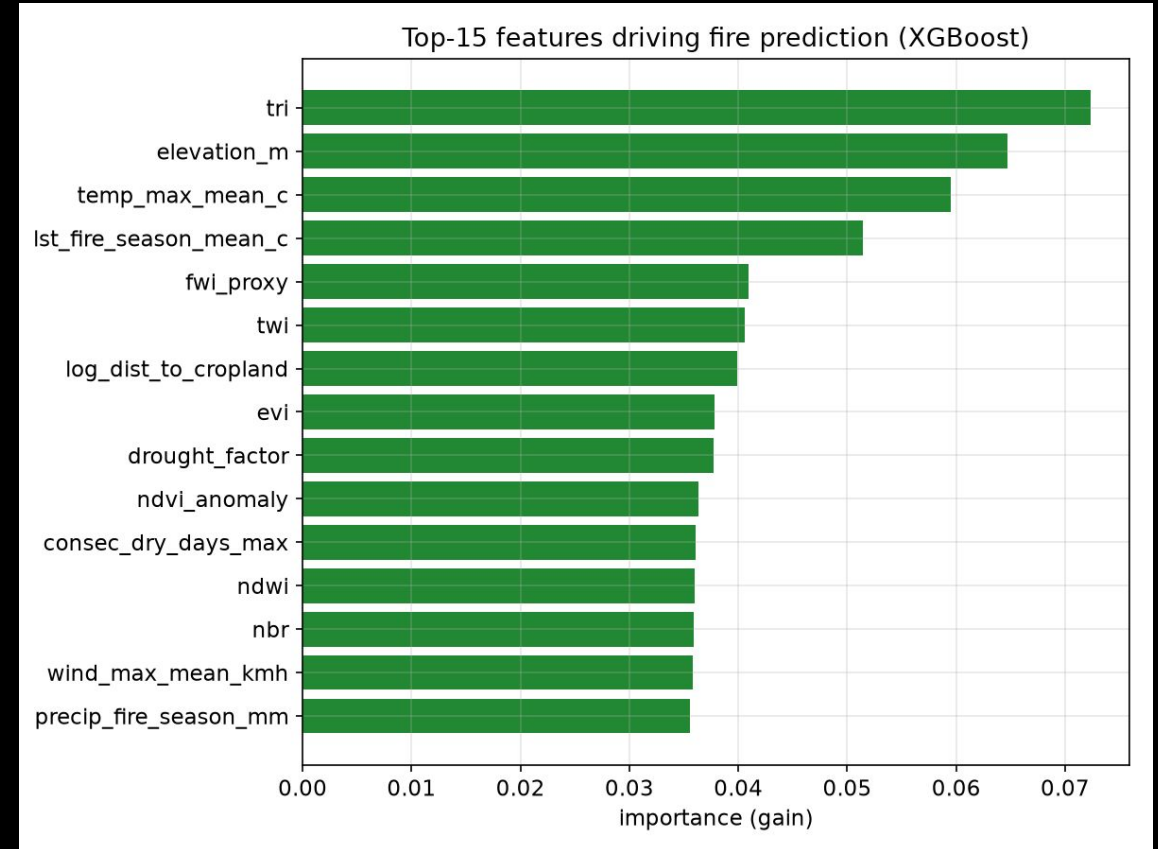
Confusion matrices & the threshold trade-off



- At **F1-optimal** threshold: catch a meaningful share of fires
- e.g. sklearn MLP: **TP 66/727**, FP 2,651 of ~442k safe cells
- **Right trade-off for a screen:** over-flagging is cheap; missing a fire is costly

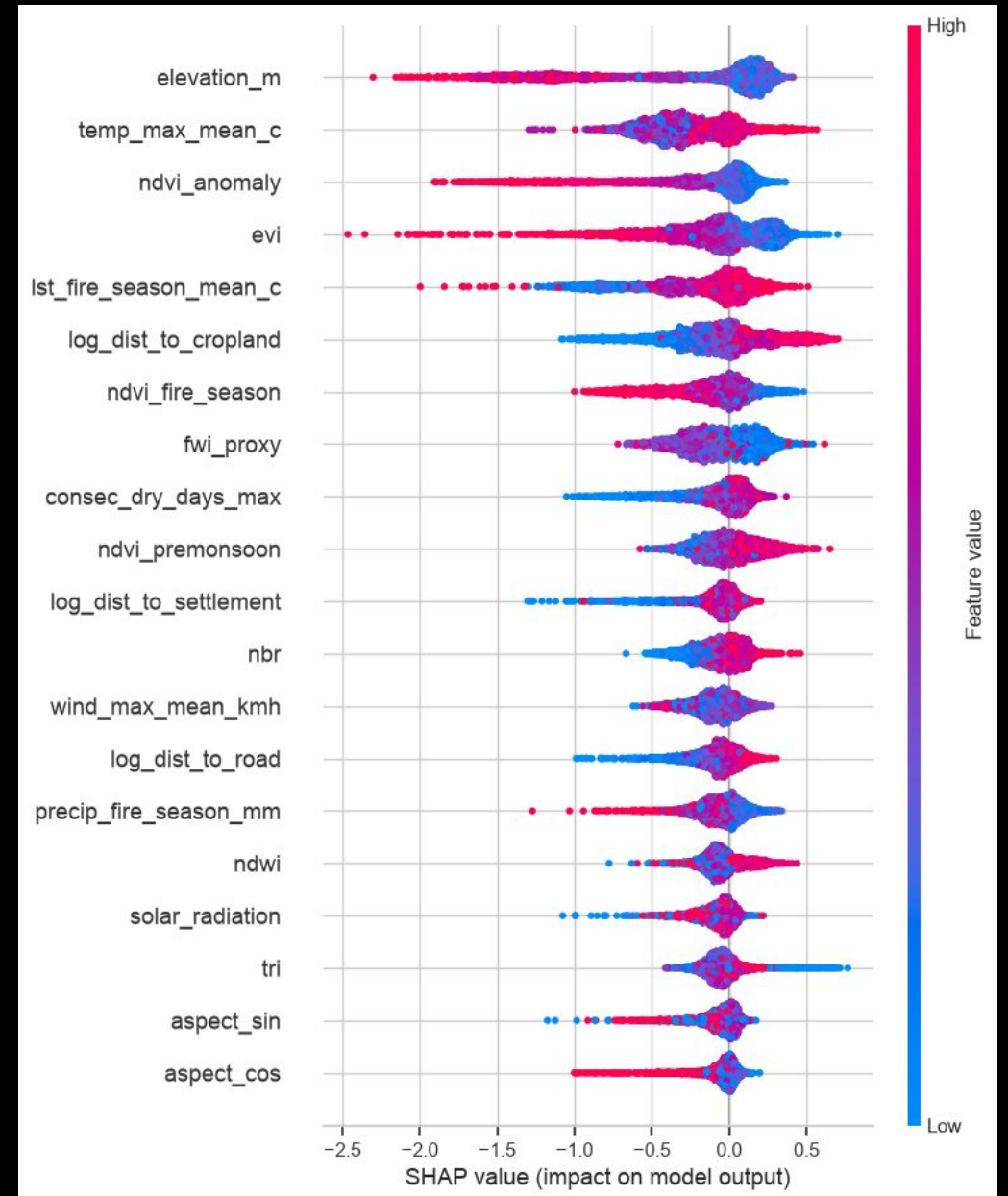
What drives the prediction

- Top drivers: **NDVI, LST, max temp, dist-to-road, drought, FWI-proxy**
- Aligns with fire ecology: dry fuel + hot/dry weather + human ignition
- SHAP confirms direction (low NDVI & short road-distance → fire)
- → gives the map **face validity**



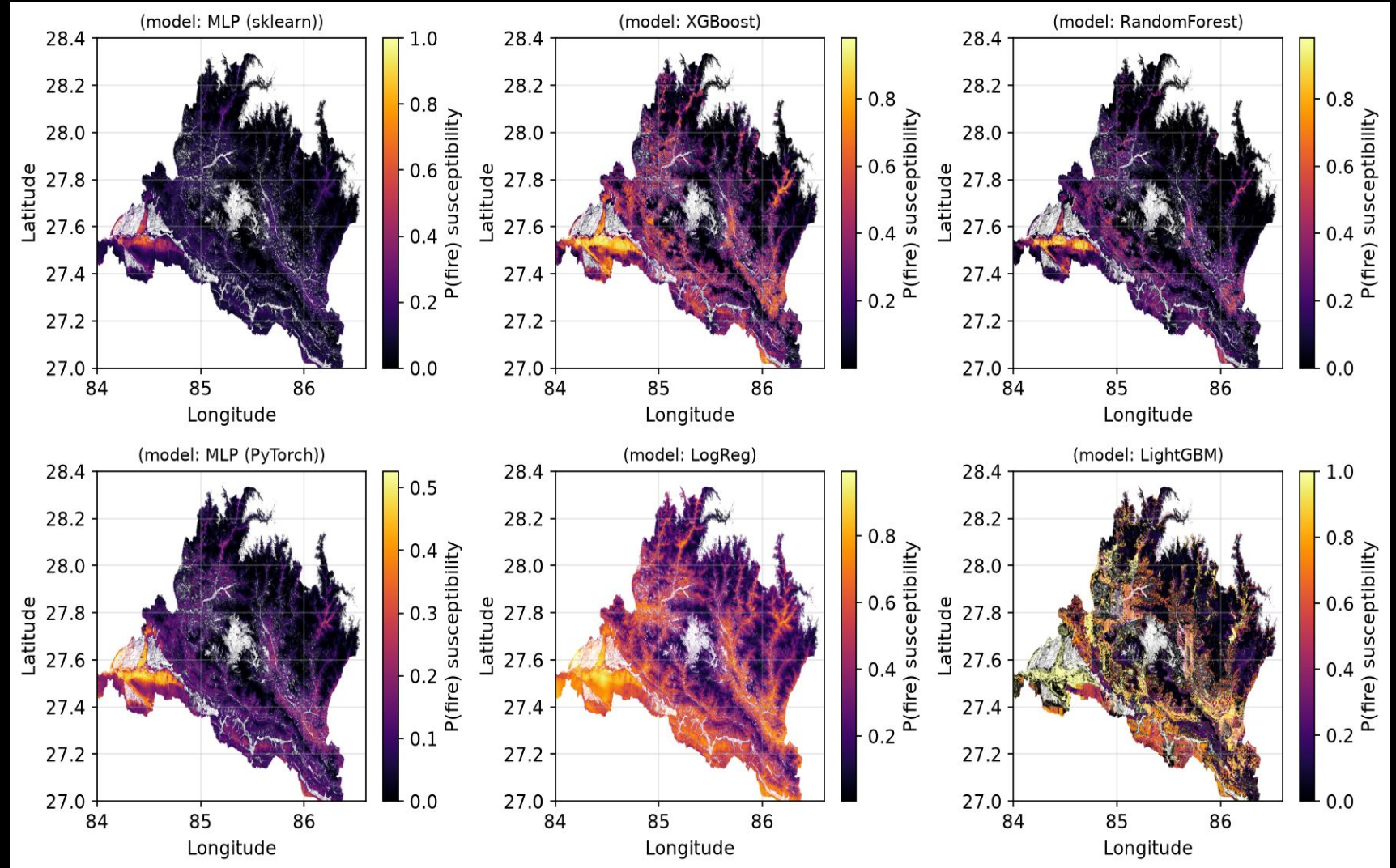
SHAP Interpretability

- SHAP applied to **XGBoost** (best tree-based model)
- Top predictors identified:
 - **slope_deg**, ndvi_premonsoon, lst_fire_season, dist_to_road
- High slope + dry veg + hot LST + near road → **high fire risk**



Susceptibility Map Output

- Model applied to all **2.1M flammable cells**
- Continuous probability raster saved as GeoTIFF



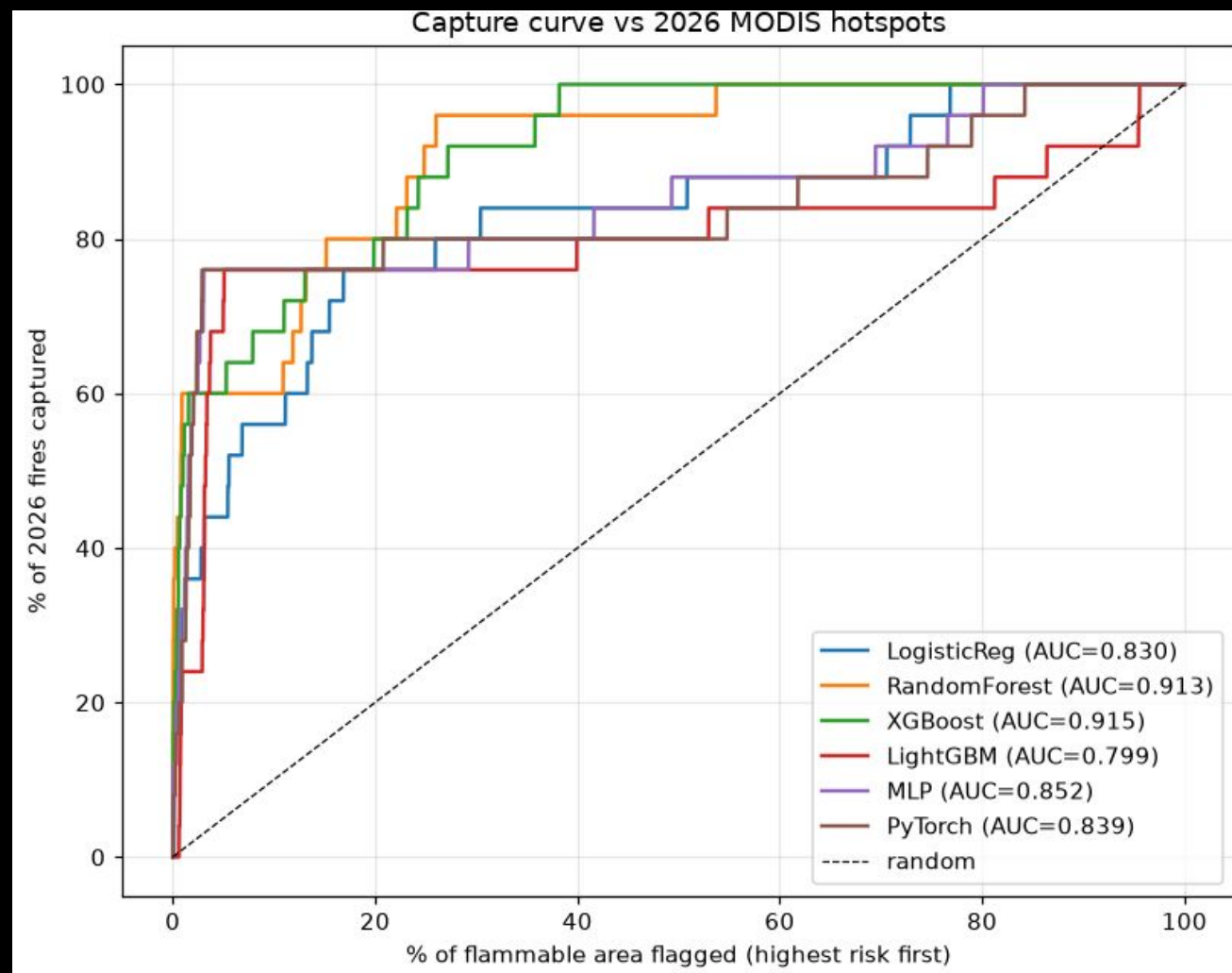
2026 Validation

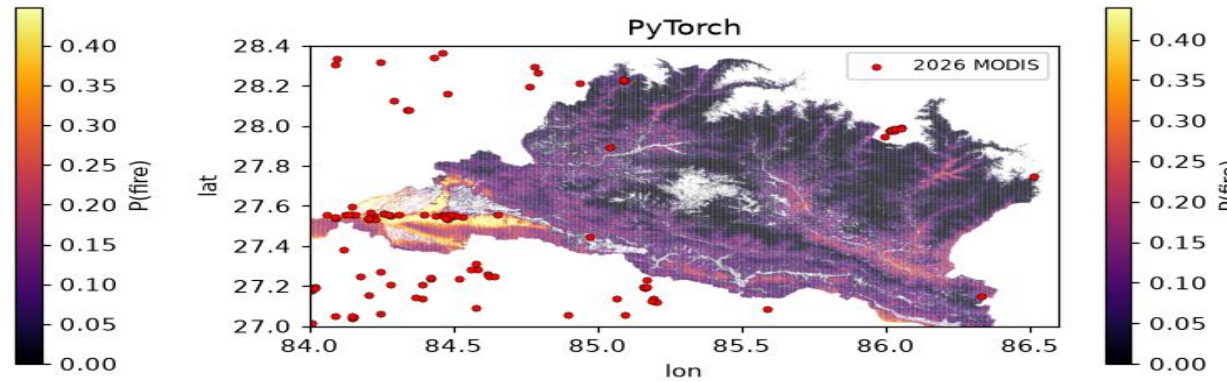
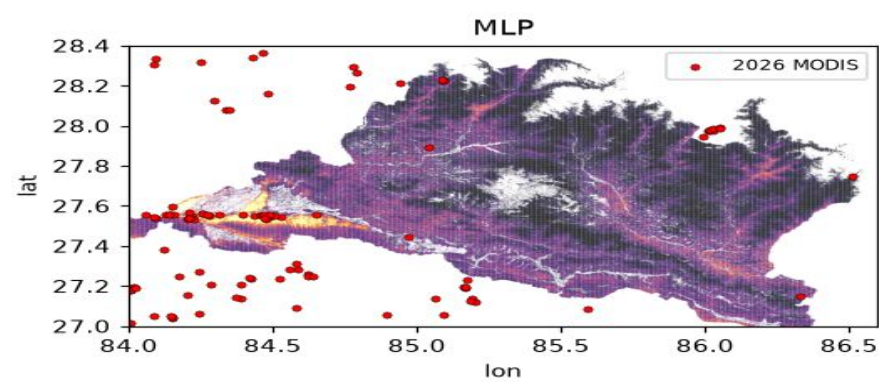
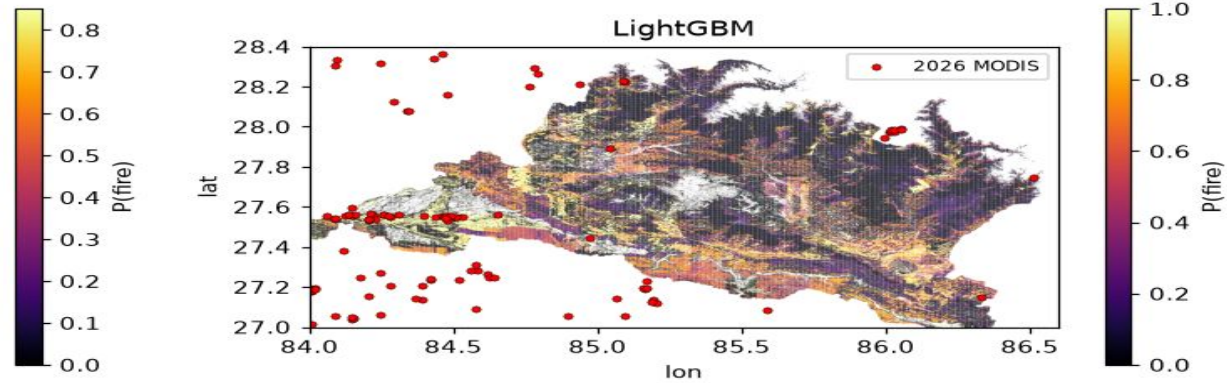
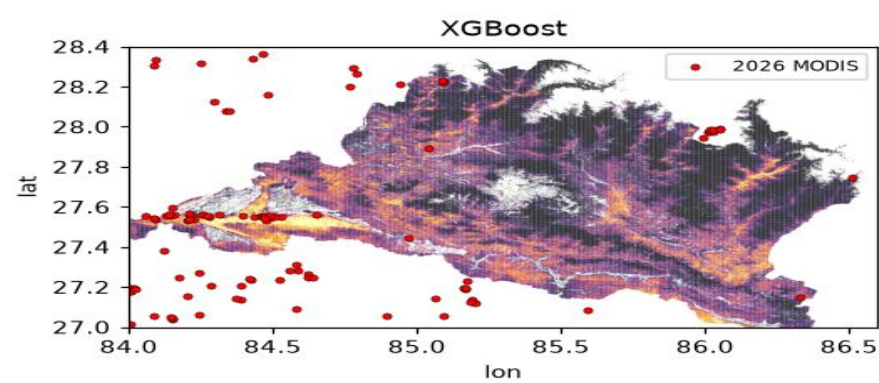
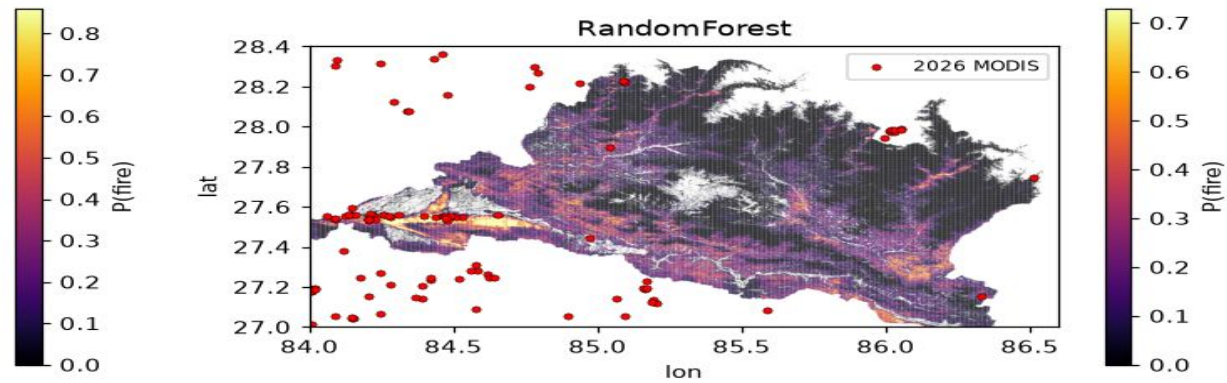
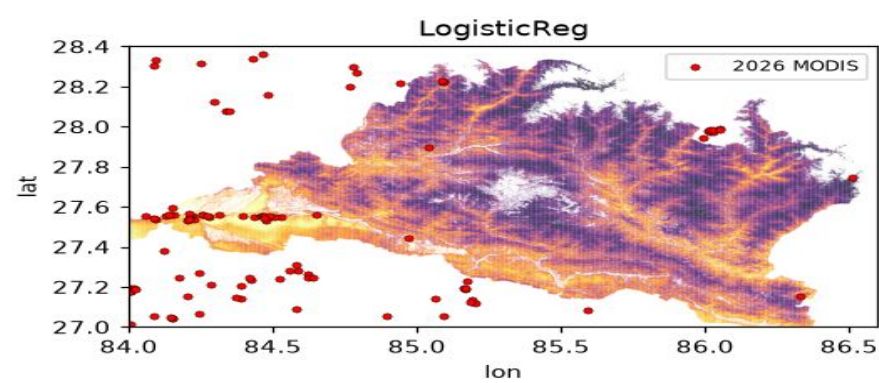
2026 Validation Design

- 2026 FIRMS hotspots **never seen** during any stage of training
- Two ground truth definitions:
 - MODIS-only: 25 confirmed flammable fire cells (conservative)
 - All-sensors (MODIS + VIIRS): 614 flammable fire cells
- Data covers **Jan–Mar 2026 only** (partial season)
 - Peak April–May not yet validated
- Key metrics: AUC–ROC, **lift over prevalence**, capture curve

2026 Results: AUC-ROC

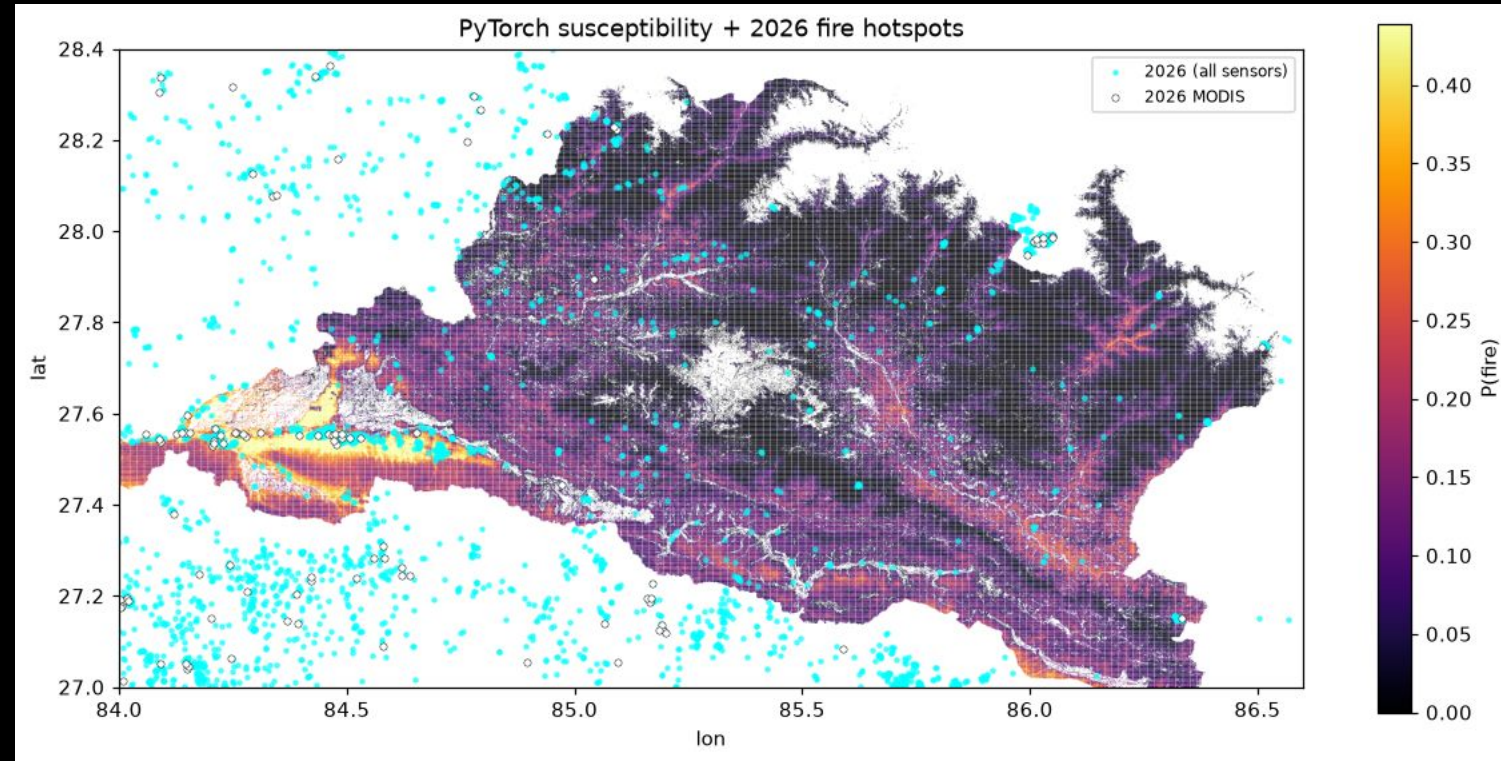
- All models achieve strong **AUC-ROC** on unseen 2026 data:
 - Random Forest: 0.913
 - XGBoost: 0.915
 - sklearn MLP: 0.852
 - PyTorch MLP: 0.839
 - Logistic Regression: 0.830
 - LightGBM: 0.798
- Models consistently **rank fire above non-fire** on future data
- RF + XGBoost lead – consistent with training holdout performance





2026 Results: Risk Class Capture

- 76% of 2026 fires fell in High or Very High risk cells
- Very High (1.7% of area) captured 52% of 2026 fires
- Top 10% of risk area captures 60–76% of 2026 fires
 - 6–7.6x more efficient than random area selection
- 16 of 25 MODIS fires were on never-burned cells (2015–2025)
 - LightGBM captures 75% of genuinely new fires in top 10% area
- Model learned susceptibility, not just fire recurrence



2026 fires by risk class (MLP)

Risk class	% of area	% of 2026 fires	Burn rate
Low	52.2%	16%	0.0004%
Moderate	35.8%	8%	0.0003%
High	10.3%	24%	0.0027%
Very High	1.7%	52%	0.0354%

→ 76% of unseen 2026 fires fell in High/Very-High cells; the Very-High class (1.7% of area) burned at ~88x the rate of the Low class.

Conclusion

Key Findings

- Extreme imbalance (513:1) – AUC-PR, not accuracy
- Spatial CV is essential – random splits inflate scores artificially
- Top predictors: slope, pre-monsoon NDVI, LST, proximity to roads
- Aligns with recent Nepal-specific literature (IOP 2025)
- MLP best on training holdout; XGBoost + RF best on 2026 future validation
- 76% of real 2026 fires in top 12% of high-risk area
- Map generalizes to genuinely new locations never burned before

Limitations & Future Work

- **FIRMS detection bias** – low-intensity understorey fires missed
- **Label noise** at cropland/forest boundaries at 100m resolution
- **Sparse records** in high-elevation Himalayan zones
- 2026 validation is **partial season** (Jan–Mar; peak Apr–May pending)
- **VPD** not yet added as a direct feature
 - Strongest Nepal predictor (2025) – currently only captured via proxies

Any Questions ?

Thank you